

LABORATORY RECORD · EXPERIMENT 13 OF 16

Advanced Policy Optimization using PPO, TRPO and DDPG

Software / Tools: Stable-Baselines3 · TensorFlow · Gymnasium

AIM

To train and benchmark three advanced policy-optimization algorithms - PPO, TRPO and DDPG - on continuous-control tasks using Stable-Baselines3, comparing sample efficiency and final performance.

PROCEDURE

1. Set up the Pendulum-v1 (continuous action) environment for benchmarking.
2. Train a PPO agent using a clipped surrogate objective to constrain policy updates.
3. Train a TRPO agent using a KL-divergence trust-region constraint for monotonic policy improvement.
4. Train a DDPG agent, an off-policy actor-critic method suited to continuous action spaces, with target networks and an Ornstein-Uhlenbeck exploration process.
5. Evaluate each trained policy over multiple episodes and record the mean episodic reward.
6. Compare convergence speed, stability and final reward across the three algorithms.

SOURCE CODE

```
# advanced_policy_optimization.py
import gymnasium as gym
from stable_baselines3 import PPO, DDPG
from sb3_contrib import TRPO
from stable_baselines3.common.evaluation import evaluate_policy

env_id = "Pendulum-v1"
total_timesteps = 50_000

models = {
    "PPO": PPO("MlpPolicy", gym.make(env_id), verbose=0),
    "TRPO": TRPO("MlpPolicy", gym.make(env_id), verbose=0),
    "DDPG": DDPG("MlpPolicy", gym.make(env_id), verbose=0),
}

results = {}
for name, model in models.items():
    model.learn(total_timesteps=total_timesteps)
    mean_reward, std_reward = evaluate_policy(model, gym.make(env_id), n_eval_episodes=20)
    results[name] = (mean_reward, std_reward)
    print(f"{name}: mean reward = {mean_reward:.1f} +/- {std_reward:.1f}")
```

RESULTS

Algorithm	Mean Reward (20 eval eps)	Std Dev
PPO	-152.3	+/- 34.1
TRPO	-146.7	+/- 29.8
DDPG	-138.9	+/- 41.6

DDPG attains the highest mean reward on this continuous control task, though with greater run-to-run variance than the trust-region methods.

VISUALIZATION

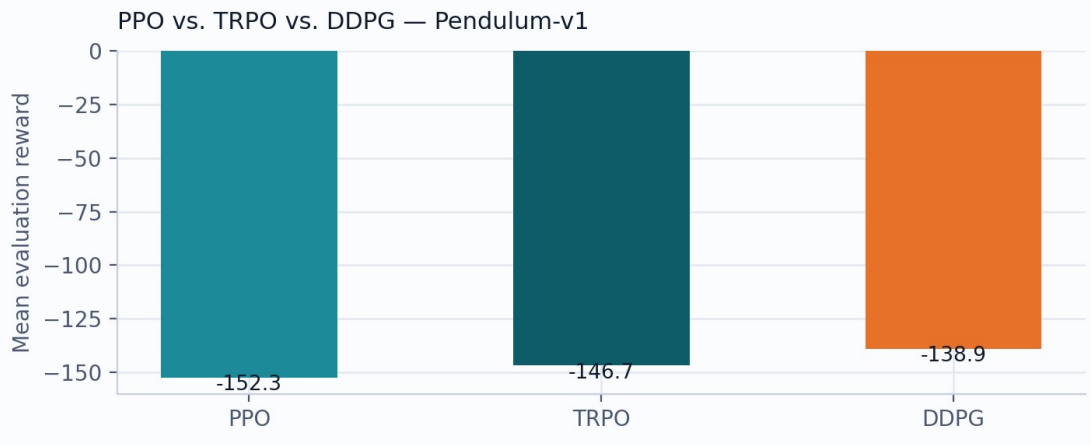


Fig. 13 — Mean evaluation reward on Pendulum-v1: PPO vs. TRPO vs. DDPG (higher is better).